

PURBANCHAL UNIVERSITY



DEPARTMENT OF COMPUTER ENGINEERING KHWOPA ENGINEERING COLLEGE LIBALI-08, BHAKTAPUR

A FINAL-DEFENSE REPORT ON **Persona Analyzer for Hiring (PAH)**

*Project work submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Computer Engineering(Third Semester)*

Submitted by

Anjila Shrestha	(810304)
Bijen Lawoju	(810307)
Nirjala Suwal	(810321)
Sanish Thapa Shrestha	(810341)

UNDER THE SUPERVISION OF
Er. Avijit Karn

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With regards,

Anjila Shrestha
Bijen Lawoju
Nirjala Suwal
Sanish Thapa Shrestha

Abstract

Persona Analyzer for Hiring (PAH) is a console-based assessment and evaluation system developed using C/C++. The system is designed to provide a structured and data-driven approach for analyzing the behavioral, psychological, and professional traits of candidates during the recruitment process. It features a secure Administrator module that allows authorized users to add, modify, and remove role-specific assessment questions, view detailed candidate reports including trait-wise scores, behavioral patterns, and suitability insights, and manage historical records through file-based storage. Additionally, the system provides a Candidate interface for registration and completion of time-limited personality assessments. Using logical scoring mechanisms and honesty detection techniques, the system generates meaningful behavioral analysis based on the Big Five personality traits, helping organizations support informed hiring and evaluation decisions.

Keywords: *Recruitment, Personality Assessment, Big Five.*

Table of Contents

Acknowledgement	i
Abstract	ii
List of Abbreviations	vii
1 Introduction	1
1.1 Background	1
1.2 Motivation	2
1.3 Problem Statement	2
1.4 Objective	2
1.5 Features of PAH	3
1.6 Scope and Applications	3
2 Literature Review	4
3 Project Management	6
3.1 Team Members	6
3.2 Feasibility Study[7]	6
3.3 Work Breakdown Planning	7
4 Methodology	8
4.1 Module Description	8
4.1.1 Add Question	8
4.1.2 View Questions	8
4.1.3 Delete Question	8
4.1.4 Calculate all candidate scores	8
4.1.5 View result by candidate ID	8
4.1.6 Logout	9
4.1.7 Candidate Panel	9
4.1.8 Overall OOP Design in Program	9
4.2 Block Diagram	10
4.3 Algorithm	11
4.4 Flowchart	12
4.5 Use Case Diagram	13
4.6 Class Diagram	14
4.7 System Sequence Diagram	15
4.8 Activity Diagram	16
4.9 Algorithms Used in the System	17
4.9.1 Advanced Sorting (Quick Sort & Numerical ID Parsing)	17
4.9.2 Stratified Randomization (Mersenne Twister)	17
4.9.3 Bitwise Data Security (XOR Cipher)	17
4.9.4 Non-Blocking Input with Timed Polling (_kbhit & Sleep)	18
4.9.5 Priority-Based Mark Assignment	18
4.9.6 Candidate ID Auto-Generation (Persistent File Counter)	18
4.9.7 BGI Bar Graph Rendering (Scaled Pixel Mapping)	19

4.9.8	Candidate File Parsing (Line-Prefix State Machine)	19
4.10	Instructions to the Candidate	20
4.11	Tools and Platforms	21
5	RESULT AND DISCUSSION	22
5.1	Home Page	22
5.2	Admin Login	23
5.3	Admin Panel	24
5.3.1	Trait Selection System	25
5.3.2	Add Questions Module	26
5.3.3	Store Questions in File	27
5.3.4	View Questions	28
5.3.5	Delete Questions	29
5.3.6	Calaculate All candidate scores	30
5.3.7	View result by candidate ID	31
5.4	Candidate Panel	33
5.4.1	Candidate Test	34
6	CONCLUSION	35
7	FUTURE ENHANCEMENT	36
	References	37

List of Figures

4.1	Block Diagram	10
4.2	Flowchart Diagram	12
4.3	Use Case Diagram	13
4.4	Class Diagram	14
4.5	System Sequence Diagram	15
4.6	Activity Diagram	16
5.1	Home Page	22
5.2	Admin Login	23
5.3	Admin Panel	24
5.4	Trait Selection	25
5.5	Add Question Module	26
5.6	Questions stored in file	27
5.7	View Questions	28
5.8	Delete Questions	29
5.9	View Questions	30
5.10	Select By Id	31
5.11	Detail Analysis	32
5.12	Candidate preview	33
5.13	Candidate giving test	34

List of Tables

3.1	Project Schedule	7
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List of Abbreviations

1. **HEXACO** – Honesty–Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, Openness to Experience
 - (a) Honesty–Humility
 - (b) Emotionality
 - (c) Extraversion
 - (d) Agreeableness
 - (e) Conscientiousness
 - (f) Openness to Experience
2. **MBTI** – Myers-Briggs Type Indicator
 - (a) Myers
 - (b) Briggs
 - (c) Type
 - (d) Indicator
3. **OCEAN** – Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism
 - (a) Openness
 - (b) Conscientiousness
 - (c) Extraversion
 - (d) Agreeableness
 - (e) Neuroticism
4. **PAH** – Persona Analyzer for Hiring
 - (a) Persona
 - (b) Analyzer
 - (c) Hiring

Chapter 1

Introduction

1.1 Background

The Persona Analyzer for Hiring (PAH) is a console-based assessment and evaluation system developed using C++. It is designed specifically for the military domain to support recruitment and selection processes by providing a structured, data-driven method to evaluate the behavioral, psychological, and professional traits of candidates applying for various military roles.

In traditional military recruitment, even after multiple interview and physical test rounds, commanders and HR personnel often face difficulty in identifying the most suitable candidates from several qualified applicants. Critical traits such as discipline, reliability, emotional stability, decision-making under pressure, teamwork and communication cannot always be accurately judged through interviews alone. PAH aims to bridge this gap by introducing a standardized and psychologically grounded assessment platform.

The system operates with two primary user roles: Administrator and Candidate. The Administrator (Recruitment Officer/Commanding Officer) has exclusive access to the system. After secure login and verification, the admin can add, modify, or remove assessment questions tailored to military roles and operational requirements. Once candidates complete the assessment, the administrator can view detailed results, including trait-wise scores, behavior patterns, strengths, weaknesses, risk factors, and overall suitability insights. The system also maintains historical candidate reports using file handling for persistence and future reference.

The Candidate interface is designed solely for assessment purposes. Candidates provide basic details and answer structured personality questions, which include Likert-scale, ranking, and situational items. Candidates do not have access to view or modify results, ensuring confidentiality and minimizing response manipulation. Responses are automatically analyzed using logical scoring, consistency checks, and honesty detection mechanisms, converting them into meaningful behavioral insights rather than simple numerical scores.

All data is stored using file-based storage, maintaining system lightweight performance without the need for complex databases. With its user-friendly interface, secure workflow, the Persona Analyzer for Hiring helps military organizations make fair, evidence-based, and effective recruitment decisions, ensuring that selected candidates are genuinely capable and suitable for high-pressure military roles.

1.2 Motivation

The motivation behind this project is to enhance the existing hiring process by integrating a structured behavioral evaluation system and support more consistent and informed hiring decision. Traditionally, assessing a candidate's discipline, consistency, emotional stability, and overall compatibility with an organization requires extensive observation and human interpretation. Through this project, these evaluations can be digitized and analyzed systematically, allowing decision-makers to compare candidates more effectively using percentage-based trait analysis. Additionally, this project will enable us to challenge the knowledge we have gained at our class and provide us an opportunity to strengthen them, and learn something new on the way.

1.3 Problem Statement

Traditional military selection processes rely heavily on physical tests, interviews, and generalized psychological evaluations, which often lack flexibility and fail to objectively assess role-specific behavioral traits. Existing assessment tools are mostly generic, do not allow customization of questions based on military roles, and may expose results to candidates, compromising confidentiality and strategic evaluation. The Persona Analyzer for Hiring (PAH) addresses these limitations by providing a secure, administrator-controlled system that allows authorized officers to design role-specific questionnaires, securely store candidate responses, and evaluate candidates using the Big Five personality traits Openness, Conscientiousness, Extraversion, Agreeableness, and Emotional Stability (Neuroticism) through percentage-based analysis, thereby enhancing the accuracy, fairness, and effectiveness of military personnel selection.

1.4 Objective

- To support the hiring process by providing a structured and objective analysis of candidates' behavioral and personality traits.

1.5 Features of PAH

- Allows military organizations to create and modify assessment questions based on specific roles, unit requirements, and operational values.
- Evaluates traits critical for military performance, such as discipline, adaptability, decision-making under pressure, teamwork, conflict management, and logical reasoning.
- Ensures secure storage of candidate responses and assessment results with strict access control for confidentiality.
- Implements time-limited questions to test a candidate's presence of mind, reflex actions, and ability to make quick, accurate decisions in high-pressure scenarios.

1.6 Scope and Applications

- Military organizations can create and modify assessment questions tailored to specific roles, units, and operational requirements.
- Candidate responses and results are securely stored and accessible only to authorized administrators, ensuring confidentiality.
- Percentage-based evaluations of key traits such as Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism help identify the most suitable candidates.
- Provides commanders and HR teams with clear, data-driven insights into candidate capabilities, aiding recruitment and role assignment.
- Sensitive candidate information remains accessible only to authorized personnel, maintaining security and professional integrity.

Chapter 2

Literature Review

Personality assessment is crucial in military recruitment, where roles demand discipline, teamwork, emotional stability, and quick decision-making under pressure. Research shows that personality traits strongly affect performance, leadership, and adaptability in high-stress situations. As Paul Costa, one of the developers of the Big Five model, noted, "Personality traits provide a scientific and measurable way to understand human behavior across different situations." [1]

The Big Five Personality Traits model is widely used for occupational assessment, including military contexts. Traits like conscientiousness and emotional stability predict reliability and stress tolerance, while extraversion, agreeableness, and openness support teamwork, leadership, and adaptability.

The cross-cultural validity of the Big Five model has been rigorously established in the literature. McCrae and Costa (1997) examined the Five-Factor Model (FFM) across six culturally and linguistically diverse populations such as German, Portuguese, Hebrew, Chinese, Korean, and Japanese, with a combined sample of 7,134 participants. Using translations of the Revised NEO Personality Inventory (NEO-PI-R), they found that all groups produced highly similar five-factor structures, concluding that the Big Five traits are not culturally specific constructs but reflect a universal structure of human personality. This makes the model a reliable and globally applicable framework for personality assessment. [2]

Traditional assessments often use Likert-scale questions, which are simple and reliable but can be faked by candidates trying to appear ideal. To reduce this, studies suggest including situational and ranking-based questions, which evaluate real decision-making and priorities. As Ashton and Lee, the developers of the HEXACO model, highlight, "Assessments that focus on behavioral tendencies rather than self-report are more resistant to manipulation and better predict real-world outcomes." [3]

Time-limited questions are also recommended, as they test presence of mind and reflexive thinking, essential for military performance. Existing systems, however, are often generic, do not tailor questions to specific roles, and sometimes reveal results to candidates, increasing the chance of manipulated answers.

A relevant precedent for short-form personality assessment is the LifeScore Big Five Personality Test (lifescoretest.com), which measures all five OCEAN traits using a compact question set based on the International Personality Item Pool (IPIP) framework. Its scoring approach assigns a numerical value per response, producing a total score per trait — consistent with established psychometric scaling practices. The present project adopts a similar structure, using 5 questions per trait with 4 response options, yielding a maximum score of 20 per trait across all five OCEAN dimensions, enabling clear categorization of personality levels as low, moderate, or high. [4]

The proposed Persona Analyzer for Hiring (PAH) addresses these issues by using a hybrid approach: Likert, situational, and ranking questions, customized for military roles, with hidden evaluation criteria and time-limited responses. This ensures assessments are reliable, fraud-resistant, and aligned with real-world military requirements.

Chapter 3

Project Management

3.1 Team Members

The project will be carried out by the contribution of the following four team members.

- Anjila Shrestha, 3rd Sem [810304]
- Bijen Lawoju, 3rd Sem [810307]
- Nirjala Suwal, 3rd Sem [810321]
- Sanish Thapa Shrestha, 3rd Sem [810341]

3.2 Feasibility Study[7]

It is the aspect that analyzes, studies and predicts the overall success and potential of the project under several aspects. It is based on extensive research and investigation regarding the project scenario and conceptual models.

- **Economic Feasibility:** The project does not require any high financial support as it will be performed on a free platform. Further, no financial needs till its completion.
- **Operation Feasibility:** This application will run smoothly on minimum hardware specifications and does not need high requirements.
- **Technical Feasibility:** A normal literate person who can understand general English can easily handle the operation. In case of difficulty, the logs and manuals in the program will clarify it.
- **Schedule Feasibility:** The expected duration for the completion of this project is around 3 months. The project will be completed in the allotted time frame.

3.3 Work Breakdown Planning

S. N	Job Description	Duration	1st week	2nd week	3rd week	4th week	5th week	6th week
1.	Problem Identification	3 days						
2.	Analysis	4 days						
3.	Design	7 days						
4.	Coding	21 days						
5.	Implementation & testing	6 days						
6.	Documentation	45 days						

Table 3.1: Project Schedule

Chapter 4

Methodology

4.1 Module Description

4.1.1 Add Question

This module allows the admin to add new personality assessment questions to the system. The admin selects a trait from the Big Five personality model (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism), enters the question text along with four answer options, and ranks the options by priority to assign marks (4, 3, 2, 1 points).

4.1.2 View Questions

The “View Questions” option enables the administrator to display all stored questions in a clear and organized format. Before displaying, the system arranges the questions in a sorted order using Quick Sort algorithm, based on their question ID, ensuring consistency and readability.

4.1.3 Delete Question

After deletion the remaining questions are automatically renumbered so the sequence stays clean, if question 1.2 is deleted, the old 1.3 becomes the new 1.2. This prevents gaps or inconsistencies in the ID system that scoring depends on

4.1.4 Calculate all candidate scores

This reads every registered candidate’s response file, calculates their score for each of the five traits, determines whether each score falls in the LOW, MODERATE, or HIGH band, and produces a full written interpretation. All results are printed to the console and simultaneously saved to a file as a permanent record.

4.1.5 View result by candidate ID

This displays a formatted table of all registered candidates, prompts the admin to enter a candidate ID, then shows that individual’s full personality report consisting trait scores, bands, and written interpretations. It then opens a separate graphical window displaying the same scores as a color-coded bar chart for visual comparison.

4.1.6 Logout

This ends the admin session and returns the program to the main menu. The admin must re-authenticate with the password to enter the panel again. It is a deliberate security boundary where no admin actions are possible after logout without passing through the login gate once more.

4.1.7 Candidate Panel

When a candidate selects the Candidate Panel from the main menu, the assessment system initializes and presents a welcome screen explaining the test structure which consists of 25 questions, 60 seconds per question, and multiple choice format, along with instructions on how to proceed honestly and carefully.

The candidate is then asked to enter their personal details. They provide their full name, age, and phone number. The system validates each entry where age must be between 18 and 25, and the phone number must be exactly 10 digits beginning with 9. If either entry fails validation, the candidate is prompted to re-enter until a valid value is provided.

Once details are confirmed, the system generates a unique Candidate ID (e.g., C001, C002) by reading a persistent counter file, incrementing it, and formatting it with leading zeros.

The system then loads all available questions from the question bank and assembles a randomised set of 25 questions. Here five question from each of the five personality traits. The questions are shuffled so no two candidates receive them in the same order, and traits are interleaved so the candidate cannot predict which personality dimension is being measured at any point.

The assessment then begins. Questions are presented one at a time, each displaying the question text and four answer options labelled A through D. A live 60-second countdown ticks on screen for each question. The candidate types their chosen option and presses Enter to submit. If the 60 seconds expire before an answer is given, the question is automatically skipped and recorded as unanswered with zero marks. The time taken to answer each question is also recorded alongside the chosen option.

After all 25 questions are answered or timed out, the collected responses are sorted back into trait order and written to the candidate's personal response file along with their details and marks per question. The file is saved under a name combining the candidate's name and their unique ID, and an entry is added to the central candidate registry so the admin can locate and review their results later. The candidate is then shown their Candidate ID and informed that their assessment is complete.

4.1.8 Overall OOP Design in Program

The Admin Panel is built using strong object-oriented principles. **Abstraction** is achieved through base classes that define common behaviors, **inheritance** allows reuse of menu and file-handling functionality, **polymorphism** enables flexible execution of different operations through a common interface, and **encapsulation** ensures that data is securely managed within structured components.

This design makes the system modular, maintainable, and scalable.

4.2 Block Diagram

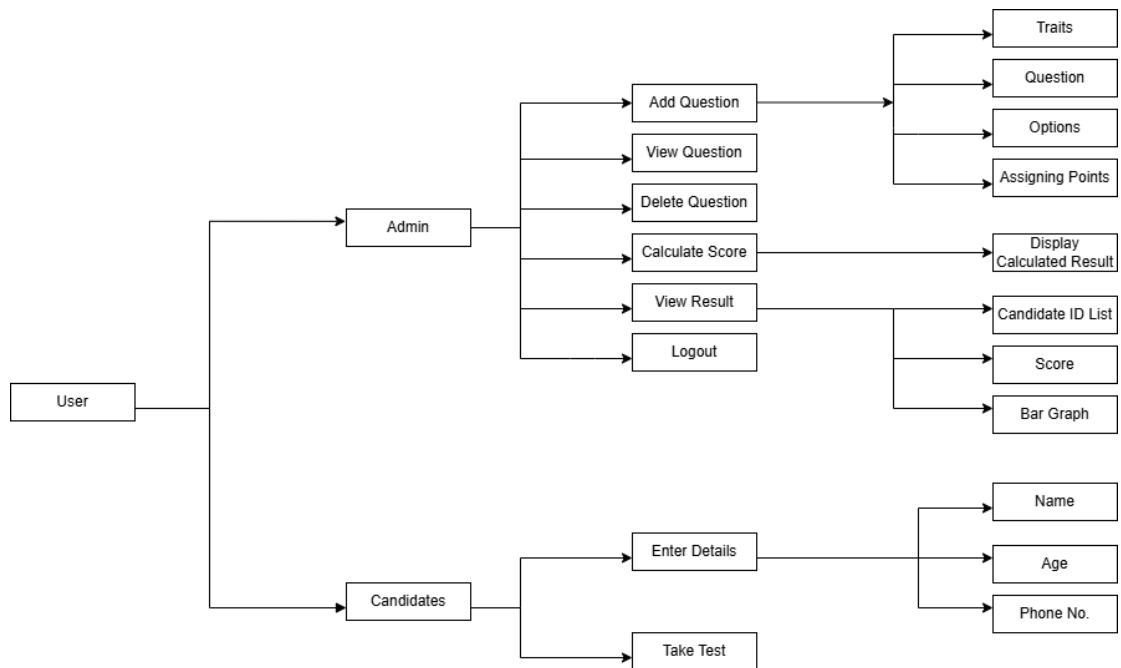


Figure 4.1: Block Diagram

4.3 Algorithm

1. Start the program.
2. Display the Main Menu:
 - Admin Login
 - Candidate Assessment
 - Exit
3. Read the user's choice.
4. If Admin Login is selected:
 - Ask for admin password.
 - Validate password with maximum 3 attempts.
 - Lock the system for 30 seconds if attempts exceed limit.
 - Display Admin Menu after successful login.
5. Perform Admin Operations:
 - Add, view, or delete questions.
 - Calculate candidate scores.
 - View candidate reports and graphs.
 - Logout and return to Main Menu.
6. If Candidate Assessment is selected:
 - Load and randomly select 25 questions.
 - Enter and validate candidate details.
 - Generate Candidate ID and response file.
 - Start timed assessment for each question.
 - Save answers, marks, and response time.
 - Mark unanswered questions as TIMEOUT.
 - Save responses and display Candidate ID.
7. If Exit is selected:
 - Display goodbye message.
 - End the program.

4.4 Flowchart

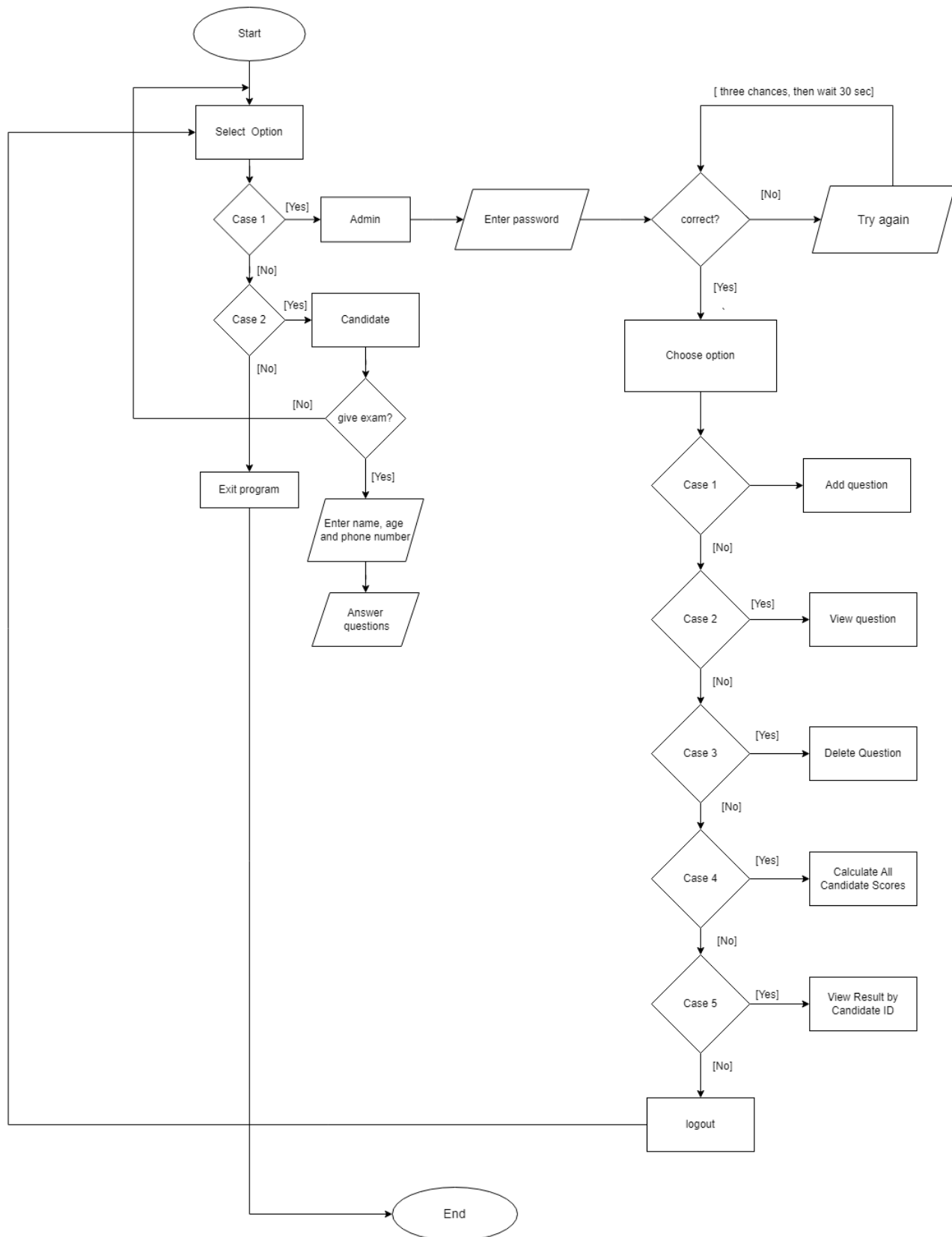


Figure 4.2: Flowchart Diagram

4.5 Use Case Diagram

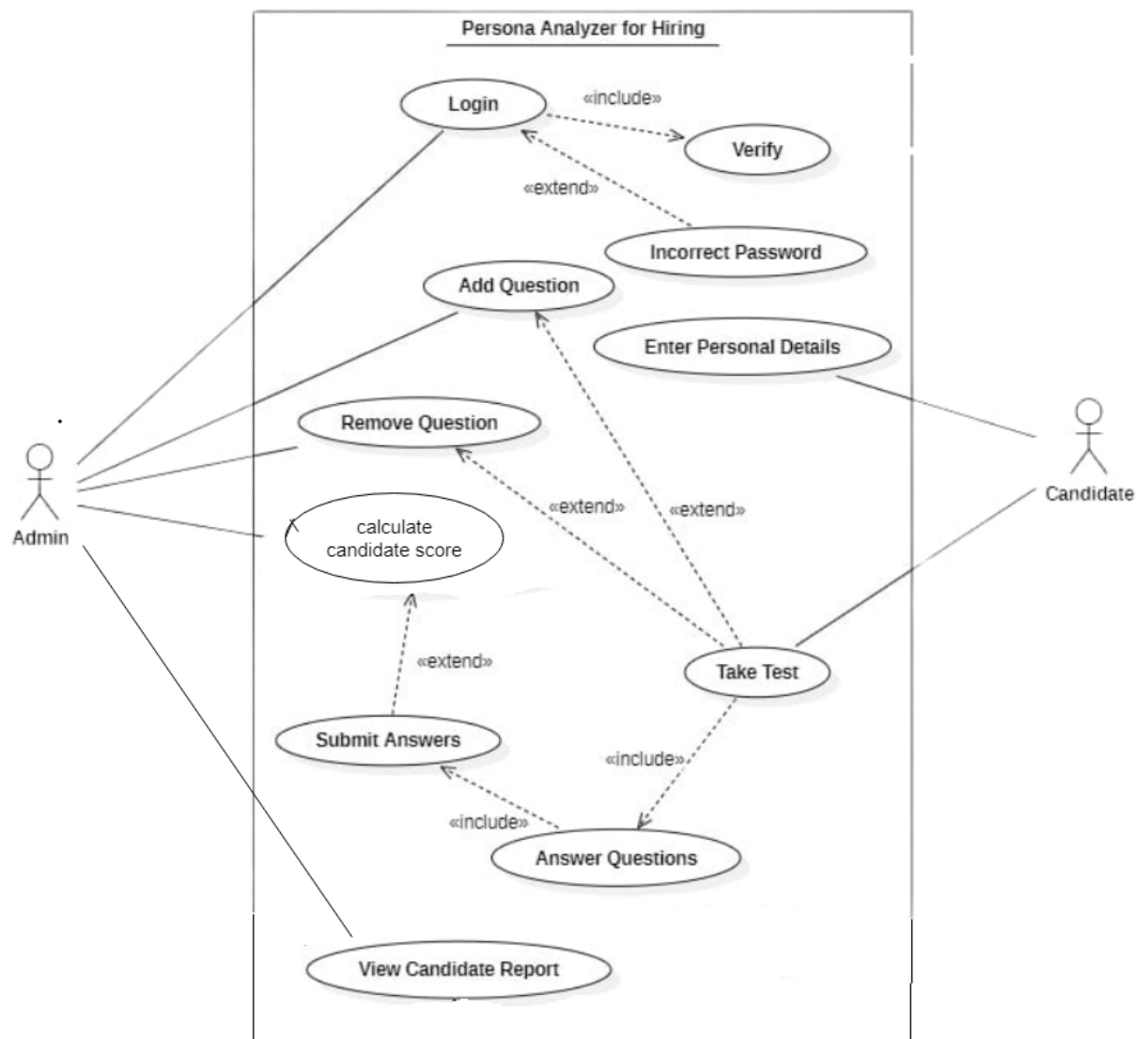


Figure 4.3: Use Case Diagram

4.6 Class Diagram

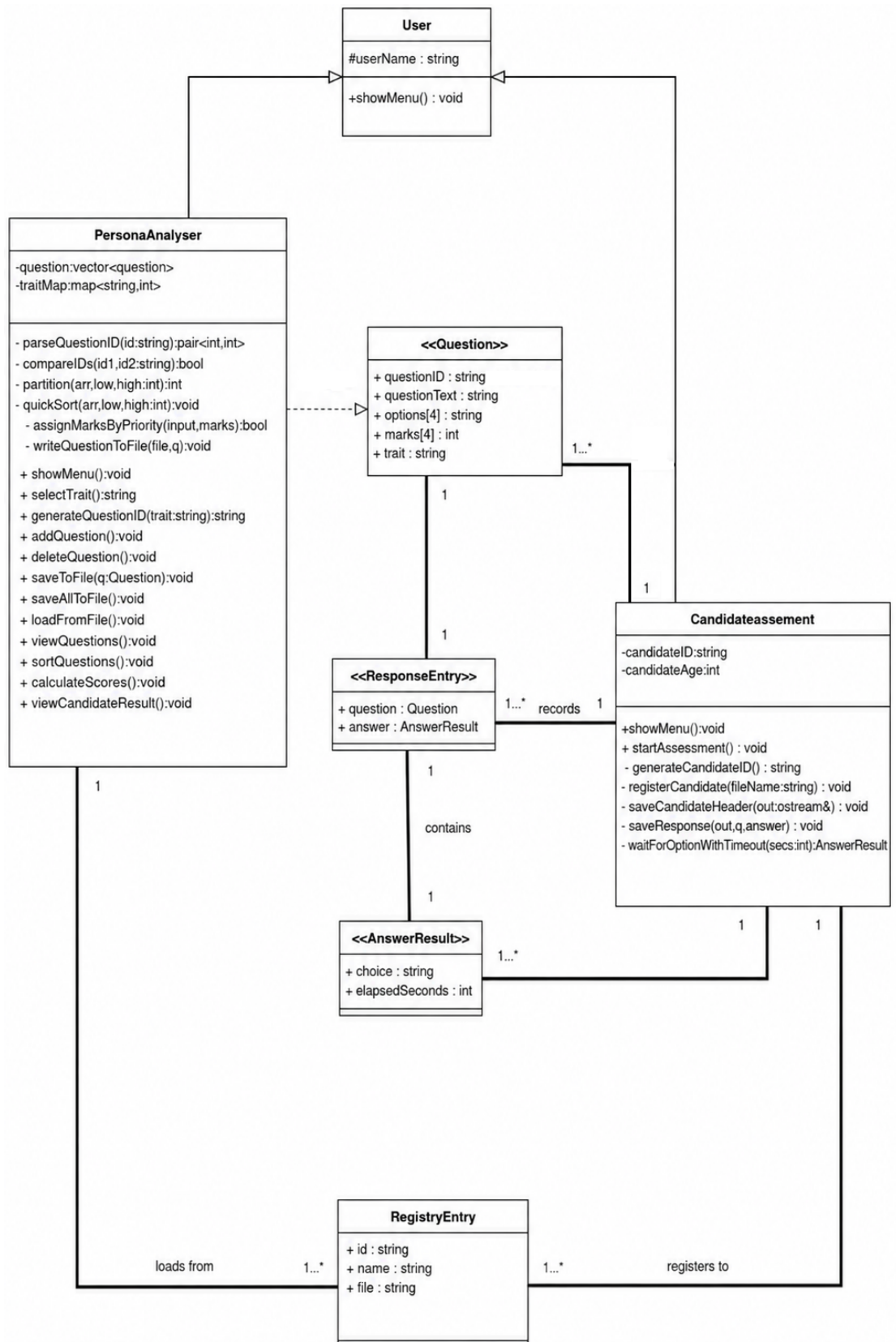


Figure 4.4: Class Diagram

4.7 System Sequence Diagram



Figure 4.5: System Sequence Diagram

4.8 Activity Diagram

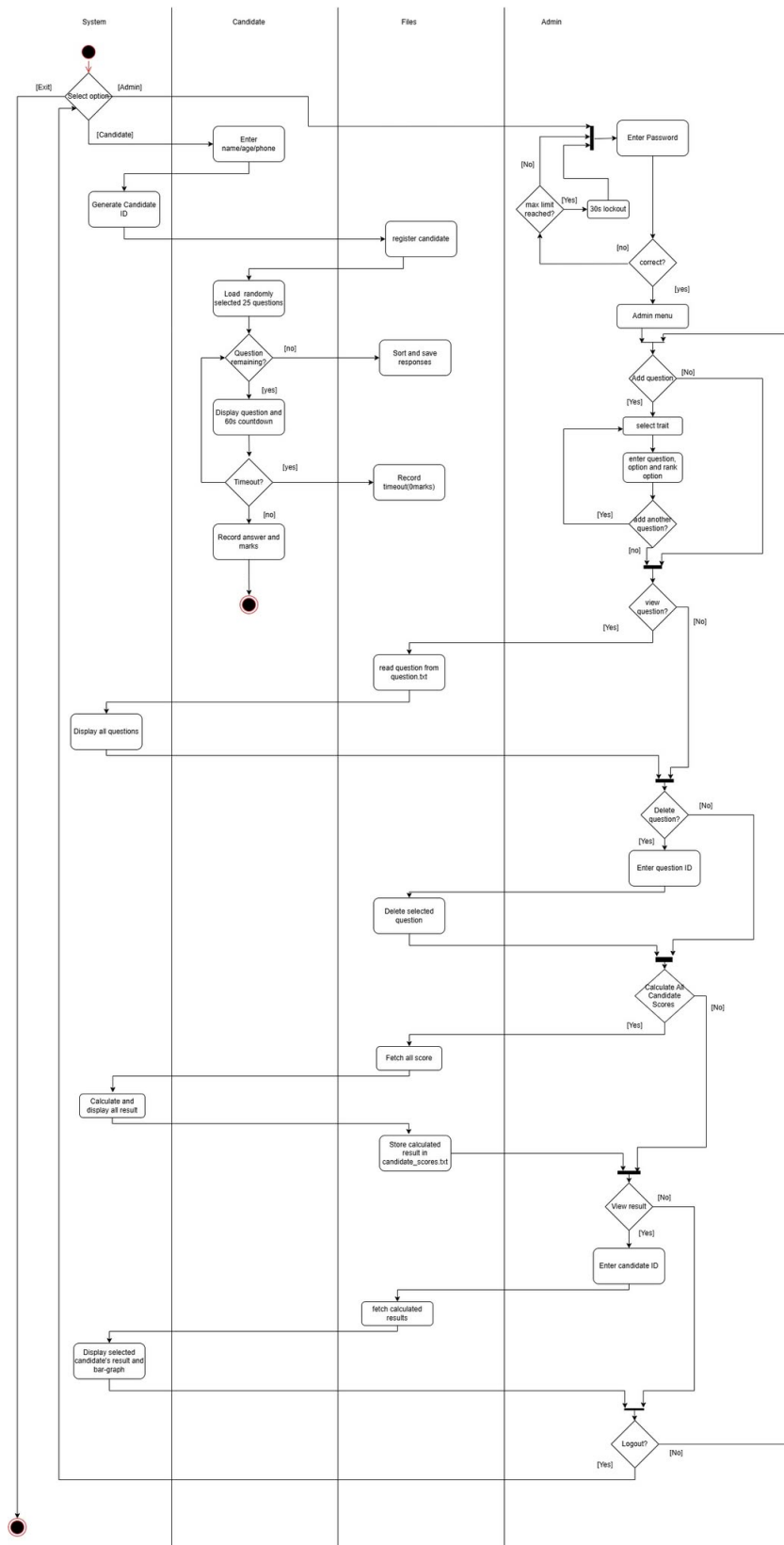


Figure 4.6: Activity Diagram

4.9 Algorithms Used in the System

The system utilizes several specialized algorithms to ensure data integrity, psychological balance, and a responsive user experience. The technical details of these implementations are outlined below.

4.9.1 Advanced Sorting (Quick Sort & Numerical ID Parsing)

Purpose: To arrange questions in true numerical order (e.g., “1.2” before “1.10”).

Logic: Uses Quick Sort with a custom comparator that parses dot-separated ID strings (e.g., “2.10”) into integer pairs using `parseQuestionID()`. It selects a pivot, partitions the data into smaller/larger segments, and recursively sorts each partition, with an average complexity of $O(n \log n)$.

Significance: Prevents lexicographic “alphabetical sorting” errors where “1.10” would incorrectly appear before “1.2”, and ensures a logical, structured question flow for the admin.

4.9.2 Stratified Randomization (Mersenne Twister)

Purpose: To generate a unique, balanced 25-question test for every candidate.

Logic: Questions are first grouped by their Big-Five trait using an `unordered_map`. The Mersenne Twister (MT19937) engine, seeded with the current high-resolution clock timestamp, then shuffles each trait group independently. The system selects exactly 5 questions per trait to guarantee stratified balance, and shuffles the final combined set a second time to randomize question order.

Significance: Guarantees every candidate receives a fair, non-repetitive, and psychometrically balanced assessment, eliminating any predictable question patterns across sessions.

4.9.3 Bitwise Data Security (XOR Cipher)

Purpose: To obfuscate the admin password and protect it from plain-text reading.

Logic: A symmetric XOR ($\oplus$) operation is applied between each character of the password string and a fixed 8-bit key (“K”). Since XOR is its own inverse, the same `encryptDecrypt()` function handles both encryption and decryption. The stored encrypted value is compared against the encrypted form of the user’s input, so the plain-text password never appears directly in any comparison.

Significance: Protects the admin credential from casual plain-text inspection without requiring any external cryptographic libraries, keeping the system self-contained.

4.9.4 Non-Blocking Input with Timed Polling (`_kbhit` & `Sleep`)

Purpose: To enforce a 60-second time limit per assessment question without freezing the program.

Logic: The `waitForOptionWithTimeout()` function uses `_kbhit()` to check whether a key has been pressed without blocking execution. If no key is detected, `Sleep(200)` yields the thread for 200 milliseconds before re-checking. A `chrono::steady_clock` deadline is computed at entry and compared on every loop iteration. When the deadline is reached, the function returns a `TIMEOUT` result automatically.

Significance: Delivers a responsive, exam-like environment where the countdown continues updating on screen and the program advances automatically on timeout, without hanging on a blocking cin call.

4.9.5 Priority-Based Mark Assignment

Purpose: To allow the admin to assign psychometric weightings to question options without manually entering numbers.

Logic: The admin inputs a ranked order of options (e.g., “A C D B”), which the `assignMarksByPriority()` function parses token by token. It validates that exactly four unique options from the set $\{A, B, C, D\}$ are provided, then assigns descending integer marks (4, 3, 2, 1) to each option in the order given. Duplicate or invalid tokens cause the input to be rejected and re-prompted.

Significance: Simplifies question creation for non-technical admins while enforcing strict data integrity, ensuring every question has a consistent and valid mark distribution.

4.9.6 Candidate ID Auto-Generation (Persistent File Counter)

Purpose: To assign each candidate a unique, sequential, human-readable ID (e.g., C001, C002).

Logic: The `generateCandidateID()` function reads an integer counter from a persistent file `_counter.txt`, increments it, writes the new value back, and formats the result as a zero-padded string prefixed with “C”. Zero-padding is applied conditionally: two leading zeros for values below 10, one for values below 100, and none beyond that.

Significance: Guarantees uniqueness across sessions without a database, and produces clean IDs that can be used to look up candidate result files from the registry.

4.9.7 BGI Bar Graph Rendering (Scaled Pixel Mapping)

Purpose: To visually display a candidate's Big-Five trait scores as a colour-coded bar graph in a separate graphics window.

Logic: The drawCandidateBarGraph() function in drawBarGraph.h maps each trait score (range 0–20) to a pixel height using linear scaling:

$$\text{barHeight} = \left(\frac{\text{score}}{20.0} \right) \times \text{MAX_H}$$

where MAX_H is the fixed pixel distance between the X-axis and the top of the graph area (310 px). Five bars are drawn at evenly spaced centre positions using BAR_STEP = 96px, each filled with a distinct BGI colour and bordered separately. A colour legend is rendered below the graph area.

Significance: Converts raw numerical scores into an immediately interpretable visual output, allowing the admin to assess a candidate's personality profile at a glance alongside the text report.

4.9.8 Candidate File Parsing (Line-Prefix State Machine)

Purpose: To reliably extract structured candidate data and per-trait scores from unstructured text response files.

Logic: The parseCandidateFile() function reads the response file line by line and uses substr(0, N) prefix matching to identify field tags such as "CandidateID:", "Trait: ", and "Marks:". It maintains a currentTrait state variable so that when a "Marks:" line is encountered, its value is accumulated into the correct trait's running total in the traitTotals map.

Significance: Allows the system to reconstruct complete candidate results from plain-text files without any database engine, making the data portable and human-readable while still being machine-parseable.

4.10 Instructions to the Candidate

For each question, rank the 4 options from **most likely (1)** to **least likely (4)** based on how you would typically respond.

Dimension 1: Neuroticism (Emotional Stability)

1. You are assigned a task with unclear instructions and limited time.

- i. Start working while thinking about possible mistakes that might occur
- ii. Ask multiple people for reassurance before proceeding
- iii. Clarify essential points quickly and focus on execution
- iv. Delay starting until the instructions feel completely clear

Target Ranking: 3 → 1 → 2 → 4

2. A teammate repeatedly makes minor errors that affect group progress.

- i. Address the issue calmly and suggest a fix
- ii. Feel irritated but continue working without addressing it
- iii. Show visible frustration and confront the teammate
- iv. Mentally dwell on the mistake while working

Target Ranking: 1 → 2 → 4 → 3

3. You receive mixed feedback on your performance—some positive, some critical.

- i. Focus on improvement points and move forward
- ii. Feel motivated one moment and discouraged the next
- iii. Fixate on the criticism despite the positives
- iv. Lose interest in the task for the rest of the day

Target Ranking: 1 → 2 → 3 → 4

4. After speaking in a group discussion, the room becomes quiet.

- i. Assume others are processing the idea
- ii. Wonder briefly if the comment was unclear
- iii. Repeatedly think about how the comment was perceived
- iv. Feel embarrassed and withdraw from further discussion

Target Ranking: 1 → 2 → 3 → 4

5. You experience a mentally exhausting day with unfinished work.

- i. Complete the priority tasks before resting
- ii. Take a short break, then resume work
- iii. Engage in comfort activities and postpone tasks
- iv. Avoid thinking about the work entirely

Target Ranking: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

6. You are placed in a high-pressure situation with strict deadlines.

- i. Break tasks into steps and proceed steadily
- ii. Feel tense but continue working
- iii. Experience difficulty concentrating due to pressure
- iv. Feel overwhelmed and slow down significantly

Target Ranking: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

4.11 Tools and Platforms

- **Tools:**

- C and C++
- VS Code, graphics.h

- **Platforms:**

- Windows
- MacOS

Chapter 5

RESULT AND DISCUSSION

The Persona Analyzer System was developed to support recruitment and hiring by evaluating candidates based on the Big Five personality traits: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. The system provides a secure admin login to manage questions and candidate results, while candidates can register using their name, age, and phone number before taking the assessment with clearly provided instructions. All questions, results, summaries, and candidate records are stored in separate '.txt' files for organized data management. The application stores questions, candidate results, summaries, and candidate counts in separate '.txt' files for organized data management.

5.1 Home Page

The home page displays the main interface for the Persona Analyzer for Hiring (PAH) application. It features three primary options: the Admin Panel for managing candidates and system settings, the Candidate Panel for conducting assessments, and a secure Exit command.

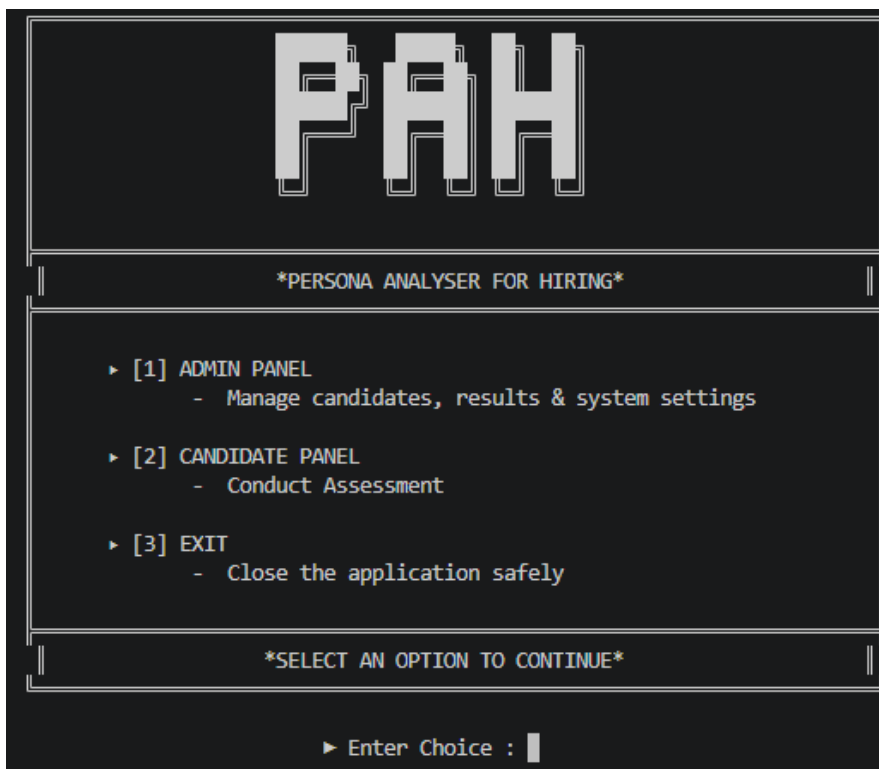


Figure 5.1: Home Page

5.2 Admin Login

The admin login system provides secure access to the admin panel. When selected, the program prompts for a password, entered in hidden mode. The password is encrypted using a simple XOR method and compared with the stored encrypted password. The admin has three attempts, after which the account is locked for 30 seconds. Correct entry grants full access to add, view, and manage questions, ensuring basic security against unauthorized access.

```
Enter Admin Password (Attempt 1/3): ***
Access Denied! Wrong Password. 2 attempts remaining.

Enter Admin Password (Attempt 2/3): **
Access Denied! Wrong Password. 1 attempts remaining.

Enter Admin Password (Attempt 3/3): ***

Access Denied! Maximum attempts exceeded.
Your account is locked. Please wait 30 seconds...
Time remaining: 1 seconds
Lockout period over. You can try again.

Enter Admin Password (Attempt 1/3): *****
Welcome, Admin! You have full access to the system.
```

Figure 5.2: Admin Login

5.3 Admin Panel

The **Admin panel** displays the specialized ****Admin Panel**** interface, which serves as the control center for assessment and data management. This menu enables administrators to maintain the question database through options to add, view, or delete individual questions. Furthermore, the panel facilitates candidate oversight by providing functions to calculate scores for all participants and retrieve specific performance metrics using a Candidate ID. Consistent with the system's command-line design, the interface includes a logout feature and a clear prompt for the user to input their numerical choice to continue.

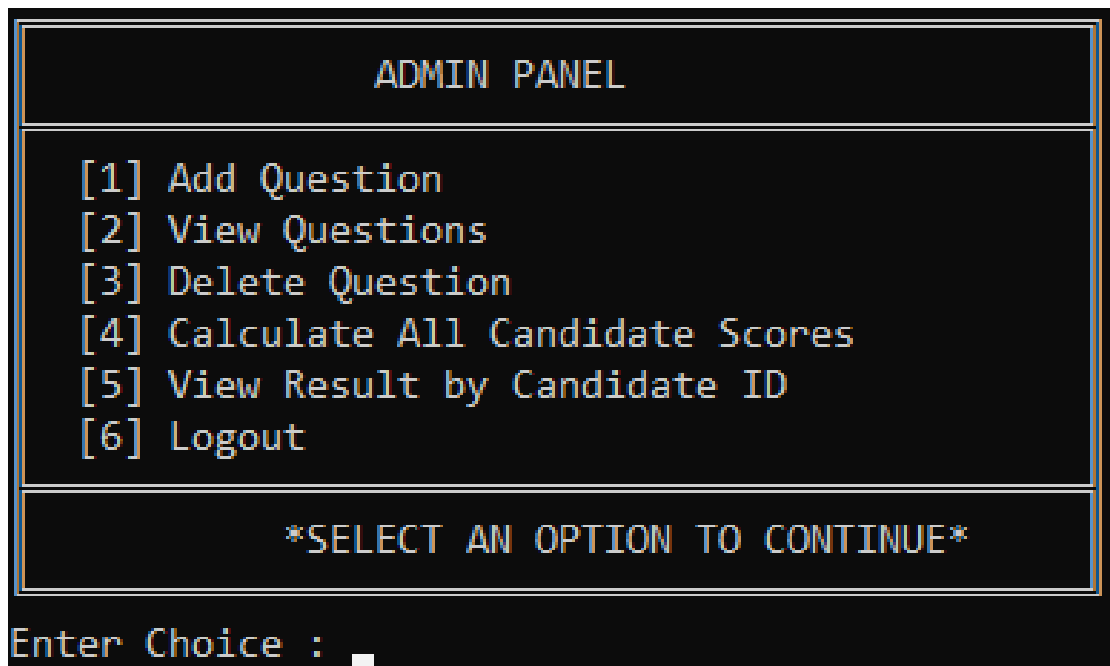


Figure 5.3: Admin Panel

5.3.1 Trait Selection System

Upon selecting the **Add Question** option from the Admin Panel menu as shown in Admin Panel, the system initiates the Trait Selection System to categorize the new question. This workflow ensures that every question added to the database is mapped to a specific personality dimension such as Openness, Conscientiousness, Extraversion, Agreeableness, or Neuroticism before it is finalized. By requiring this selection immediately after the question text is defined, the system maintains a structured data schema, allowing the automated and accurate calculation of candidate scores based on specific psychological traits during the evaluation phase.

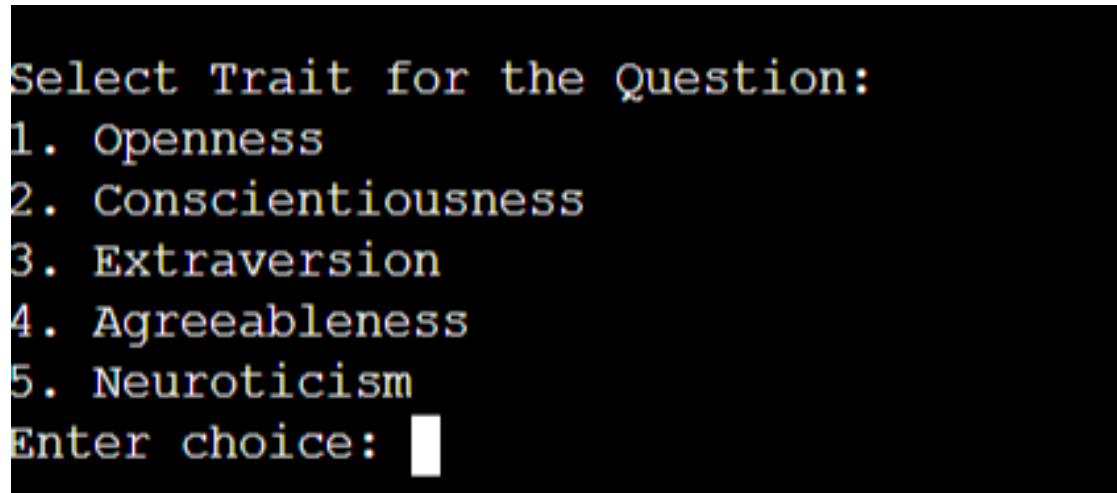


Figure 5.4: Trait Selection

5.3.2 Add Questions Module

The **Add Questions** feature lets the admin create new personality-based questions by entering the question text and four options (A–D). The admin ranks the options in order of priority, which automatically assigns marks from highest to lowest (4–1). Once entered, the question, options, and marks are saved in the system for later use.

```
Select Trait for the Question:
1. Openness
2. Conscientiousness
3. Extraversion
4. Agreeableness
5. Neuroticism
Enter choice: 5

Enter Question: you are assigned task with unclear instruction and limited time.
Enter Option A: start working while thinking about possible mistakes that might occur.
Enter Option B: ask multiple people for reassurance before proceeding.
Enter Option C: clarify essential points quickly and focus on execution.
Enter Option D: delay starting until the instruction feel completely clear.

Enter options in priority order (highest to lowest), separated by spaces.
Example: A C D B -> A=4pts, C=3pts, D=2pts, B=1pt
Your ranking: C A C B
Duplicate option 'C' found.

Enter options in priority order (highest to lowest), separated by spaces.
Example: A C D B -> A=4pts, C=3pts, D=2pts, B=1pt
Your ranking: D A C B

Marks assigned:
  Option A (start working while thinking about possible mistakes that might occur.): 3 pts
  Option B (ask multiple people for reassurance before proceeding.): 1 pts
  Option C (clarify essential points quickly and focus on execution.): 2 pts
  Option D (delay starting until the instruction feel completely clear.): 4 pts

Question Added Successfully!

Do you want to add another question? (1=Yes, 0=No):
```

Figure 5.5: Add Question Module

5.3.3 Store Questions in File

The system saves every question added by the admin, along with its four options and assigned marks, permanently in a file named questions.txt. This ensures that all data remains intact even after the program is closed, allowing the admin to view, edit, or sort questions at any time. By storing questions in a structured and organized format, the program ensures easy retrieval, efficient management, and long-term preservation of the question data, supporting smooth operation of the assessment process.

```
1 QuestionID:5.1
2 Trait: Neuroticism
3 you are assigned task with unclear instruction and limited time.
4 OptionID: 5.1A | start working while thinking about possible mistakes that might occur. | Marks: 3
5 OptionID: 5.1B | ask multiple people for reassurance before proceeding. | Marks: 1
6 OptionID: 5.1C | clarify essential points quickly and focus on execution. | Marks: 2
7 OptionID: 5.1D | delay starting until the instruction feel completely clear. | Marks: 4
8 -----
9 QuestionID:5.2
10 Trait: Neuroticism
11 a teammate repeatedly makes minor errors that affect group progress.
12 OptionID: 5.2A | address the issue calmly and suggest a fix | Marks: 1
13 OptionID: 5.2B | feel irritated but continue working without addressing it | Marks: 2
14 OptionID: 5.2C | show visible frustration and confront the teammate | Marks: 4
15 OptionID: 5.2D | mentally dwell on the mistake while working | Marks: 3
16 -----
17 QuestionID:5.3
18 Trait: Neuroticism
19 you receive mixed feedback on your performance
20 OptionID: 5.3A | focus on improvement points and move forward | Marks: 1
21 OptionID: 5.3B | feel motivated one moment and discouraged the next | Marks: 2
22 OptionID: 5.3C | fixate on the criticism despite the positives | Marks: 3
23 OptionID: 5.3D | lose interest in the task for the rest of the day | Marks: 4
24 -----
```

Figure 5.6: Questions stored in file

5.3.4 View Questions

The **View Questions** feature allows the admin to display every question stored in the system in a clear and organized format. Each question is shown along with its trait, question text, all four options, and the corresponding marks, making it easy to review and verify the stored data. This ensures that the admin can quickly access and check all questions without opening or manually reading the storage file.

```
===== PERSONA ANALYSER =====
1. Admin Panel
2. Candidate Panel
3. Exit
Enter choice: 1

===== ADMIN PANEL =====
1. Add Question
2. View Questions
3. View Specific Question
4. Back to Main Menu
Enter choice: 2

Questions sorted successfully by ID (e.g., 1.1, 1.2, 2.1, ...)!

===== Stored Questions =====

QuestionID:5.1
Trait: Neuroticism
You are assigned task with unclear instruction and limited time.
OptionID: 5.1A | start working while thinking about possible mistakes that might occur. | Marks: 3
OptionID: 5.1B | ask multiple people for reassurance before proceeding. | Marks: 1
OptionID: 5.1C | clarify essential points quickly and focus on execution. | Marks: 2
OptionID: 5.1D | delay starting until the instruction feel completely clear. | Marks: 4
-----

===== End of Questions =====
```

Figure 5.7: View Questions

5.3.5 Delete Questions

The Delete Question demonstrates the question management functionality within the system, allowing for the precise removal of entries from the assessment file. The interface presents a detailed list of current questions, each categorized by its associated personality trait and identified by a specific numerical ID. To perform a deletion, the user inputs the target ID at the prompt, after which the system confirms the removal and automatically renumbers the remaining questions to maintain a structured sequence. This feature ensures that the file remains orderly and that the underlying storage remains synchronized following any modifications.

```
ID: 4.23 | [Agreeableness] your view of your abilities compared to others
ID: 4.24 | [Agreeableness] in a conversation about achievements
ID: 4.25 | [Agreeableness] when you are clearly the expert in the room
ID: 4.26 | [Agreeableness] a colleague is visibly upset about a personal issue
ID: 4.27 | [Agreeableness] you hear about someone facing difficulties
ID: 4.28 | [Agreeableness] your response to others suffering is
ID: 4.29 | [Agreeableness] someone shares a hardship they are experiencing
ID: 4.30 | [Agreeableness] when you see someone struggling
ID: 5.1 | [Neuroticism] you are planning a trip that is three months away and thinking about preparation
ID: 5.2 | [Neuroticism] your phone battery is at 20 percent and you are out for the day
ID: 5.3 | [Neuroticism] you submit an important document and notice a small typo after submission
ID: 5.4 | [Neuroticism] your manager schedules an unexpected meeting titled quick discussion
ID: 5.5 | [Neuroticism] you are responsible for an event happening in two weeks
ID: 5.6 | [Neuroticism] someone cuts in front of you in a queue you have been waiting in for 10 minutes
ID: 5.7 | [Neuroticism] technology stops working when you are in the middle of important work
ID: 5.8 | [Neuroticism] a colleague takes credit for an idea you shared in a previous conversation
ID: 5.9 | [Neuroticism] you are explaining something for the third time to the same person
ID: 5.10 | [Neuroticism] traffic makes you late despite leaving on time
ID: 5.11 | [Neuroticism] your productivity varies significantly throughout the day
ID: 5.12 | [Neuroticism] after receiving criticism on one aspect of your work
ID: 5.13 | [Neuroticism] someone cancels plans you were looking forward to
ID: 5.14 | [Neuroticism] how would you describe your emotional experience during a typical week
ID: 5.15 | [Neuroticism] after a social interaction that felt awkward
ID: 5.16 | [Neuroticism] you are about to present in front of a group you do not know well
ID: 5.17 | [Neuroticism] you notice people whispering after you leave a conversation
ID: 5.18 | [Neuroticism] you are invited to a social gathering where you will know very few people
ID: 5.19 | [Neuroticism] after sharing an opinion in a group no one immediately responds
ID: 5.20 | [Neuroticism] you receive less interaction on a social media post than usual
ID: 5.21 | [Neuroticism] you have a free evening after a long week of work
ID: 5.22 | [Neuroticism] there is an expensive item you want but do not necessarily need
ID: 5.23 | [Neuroticism] you are trying to maintain a healthy routine but crave something indulgent
ID: 5.24 | [Neuroticism] when stressed your first instinct is to
ID: 5.25 | [Neuroticism] you are on a tight deadline but feel mentally exhausted
ID: 5.26 | [Neuroticism] multiple urgent tasks arrive simultaneously
ID: 5.27 | [Neuroticism] you are managing a project with constantly changing requirements
ID: 5.28 | [Neuroticism] during a high-stakes situation your performance
ID: 5.29 | [Neuroticism] you have back-to-back commitments with no buffer time
ID: 5.30 | [Neuroticism] when facing criticism during a stressful period

Enter the Question ID to delete (e.g. 1.3): 1.31
Question 1.31 deleted and IDs renumbered successfully.
```

Figure 5.8: Delete Questions

5.3.6 Calculate All candidate scores

The **Calculate All Candidate Scores** feature allows the System to calculate and display result of a candidate test. It stores in the system in a clear and organized format.

```
=====
      NEPAL ARMY - CHARACTER ASSESSMENT REPORT
=====
CandidateID : C002
Name       : Nirjala Suwal
Age        : 19
Phone      : 6565656
-----
TRAIT SCORES & PROFILE
-----
1. Neuroticism      : 11/20 [MODERATE]
   -> Occasionally affected by stress, generally manages emotions adequately.

2. Openness         : 12/20 [MODERATE]
   -> Adaptable when needed, balances tradition with flexibility.

3. Extraversion     : 12/20 [MODERATE]
   -> Comfortable in both team and solo tasks, situationally assertive.

4. Agreeableness    : 15/20 [MODERATE]
   -> Cooperative yet capable of standing firm when needed.

5. Conscientiousness : 15/20 [MODERATE]
   -> Reasonably organized and reliable under normal conditions.

-----
OVERALL SCORE      : 65/100
-----
TRAIT SUMMARY
  Strongest Trait   : Agreeableness (15/20)
  Weakest Trait     : Neuroticism (11/20)
  * Note: Low Neuroticism = High Emotional Stability
=====
```

Figure 5.9: View Questions

5.3.7 View result by candidate ID

The **View Specific Result of a candidate** showcases the advanced reporting capabilities of the system, which combines text-based data with a graphical representation of a candidate's assessment. Upon entering a specific Candidate ID, the application generates a detailed character assessment report alongside a separate graphical window labeled "Trait Bar Graph". This visualization utilizes the BGI library to render a color-coded bar chart, where each bar corresponds to one of the Big Five personality traits: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. By presenting these scores on a scale of 0 to 20, the system allows evaluators to perform a rapid comparative analysis of a candidate's psychological dimensions, effectively bridging low-level graphics with high-level data analysis for clearer decision-making.

```
ADMIN PANEL

[1] Add Question
[2] View Questions
[3] Delete Question
[4] Calculate All Candidate Scores
[5] View Result by Candidate ID
[6] Logout

*SELECT AN OPTION TO CONTINUE*

Enter Choice : 5

+-----+-----+
| ID      | Name                               |
+-----+-----+
| C001    | Sanish Thapa Shrestha             |
| C002    | Nirjala Suwal                     |
| C003    | Bijen Lawoju                      |
| C004    | Anjila Shrestha                   |
+-----+-----+

Enter Candidate ID:
```

Figure 5.10: Select By Id

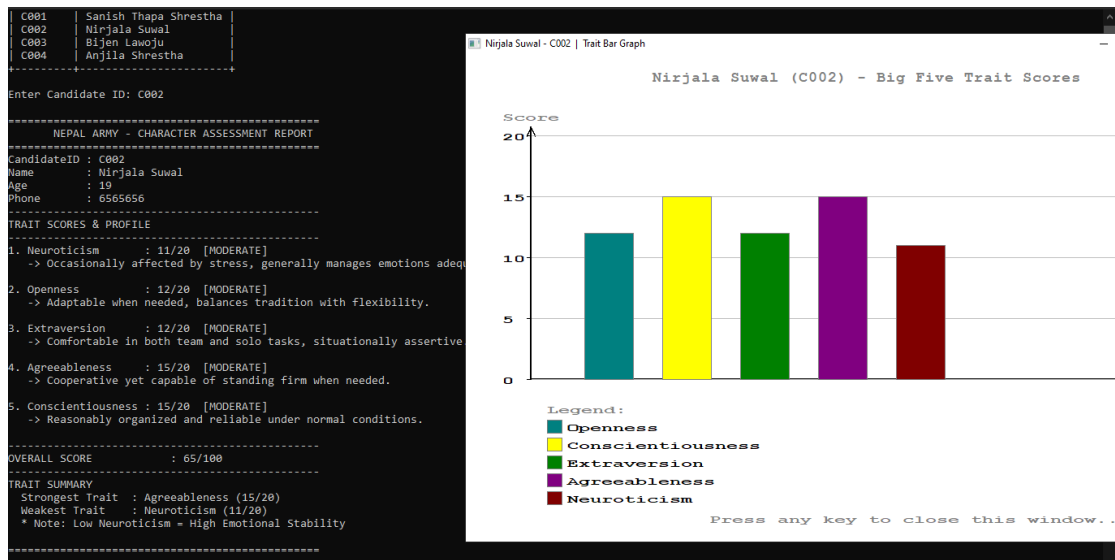


Figure 5.11: Detail Analysis

5.4 Candidate Panel

The **Candidate panel** illustrates the initial landing page for the Candidate Panel, where users are introduced to the assessment's purpose and rules. It outlines a clear "Test Structure," specifying that the assessment consists of 25 multiple-choice questions with a strict 60-second limit per question. The "Instructions" section emphasizes critical operational details, such as the inability to return to previous questions and the automatic skipping of questions if the timer expires, ensuring the candidate is fully informed before providing their details to begin.

```

  ▶ [2] CANDIDATE PANEL
      □ Conduct Assessment

  ▶ [3] EXIT
      □ Close the application safely

*SELECT AN OPTION TO CONTINUE*

▶ Enter Choice : 2

NEPAL ARMY - CHARACTER ASSESSMENT TEST

ABOUT THIS TEST
-----
This is a personality assessment which will evaluate your personality

TEST STRUCTURE
-----
- Total Questions      : 25
- Time per Question    : 60 seconds
- Total Time (max)     : ~25 minutes
- Answer Format         : Multiple Choice (A, B, C or D)

INSTRUCTIONS
-----
* Read each question carefully before answering.
* Type A, B, C, or D and press ENTER to submit your answer.
* If you do not answer within 60 seconds, the question will
  be skipped and marked as unanswered (0 marks).
* There are no right or wrong answers - answer honestly.
* You cannot go back to a previous question.
* Do not close the program during the test.

-----
When you are ready, please enter your details below to begin.
-----
```

Figure 5.12: Candidate preview

5.4.1 Candidate Test

This image showcases the system's robust data validation and live testing environment. Before the assessment begins, the program enforces strict input rules for "Candidate Details," rejecting invalid ages (outside the 18–25 range) and improper phone number formats to ensure data integrity. Once the test starts, the interface presents a single question at a time with four distinct personality-based options; notably, it features a dynamic countdown timer that tracks the remaining seconds for the current question, providing a real-time, high-pressure environment for the user to submit their response.

```
===== Candidate Details =====
Enter your name: nirjala suwal
Enter your age: 67
Invalid age. Enter age between 18 and 25.
Enter your age: 23
Enter your phone number:
Phone number cannot be empty. Please enter your phone number.
Enter your phone number: 3
Invalid phone number. Please enter a valid phone number.
Enter your phone number: 9876543210

Assessment starts now. You get 60 seconds per question.
Enter options (type A, B, C or D)

Question 1 / 25
you are planning a vacation
A. visit a new place but with planned activities
B. explore with flexible plans
C. choose a familiar type of destination
D. seek out completely novel experiences
Your answer (type A, B, C or D) – Time left: 48s:  _
```

Figure 5.13: Candidate giving test

Chapter 6

CONCLUSION

We developed the Persona Analyzer for Hiring (PAH) using concepts such as classes, files, vectors, structures etc. in C++. The system conducts personality assessments for job applicants based on 25 questions derived from the Big Five Personality Model and calculates the personality traits of each candidate. The generated reports are then displayed to the administrator for evaluation. Additionally, the administrator has the ability to add or delete questions according to specific job requirements.

Chapter 7

FUTURE ENHANCEMENT

- Replace file-based storage with a proper database (SQLite/MySQL)
- Allow batch comparison and ranking of multiple candidates
- Multiple admin logins
- Migrate to client-server architecture for remote assessments

References

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